EXPERIMENT 5

Academic Year: 2025-26 **Year and Sem: BEVIII**
Subject: Recommendation Systems Lab **Course Code: CSDOL8022**

Aim: Implementation of Bayes Classifier for recommendation system using Python
Theory: The Naive Bayes classifier is a simple probabilistic classifier based on applying Bayes' theorem with the "naive" assumption of conditional independence between features. In the context of recommendation systems, Naive Bayes can be used to predict the likelihood of a user liking or disliking an item based on a set of features or past interactions. This is particularly useful for categorical data, where we can compute the probabilities of the user liking an item based on their profile or preferences. For simplicity, let's implement a Naive Bayes classifier to make item recommendations based on user ratings or preferences. In this case, we'll use a very basic approach where users rate products, and the system uses these ratings to predict which products the user will like.
In recommendation systems, the steps to implement a Naive Bayes classifier typically involve:
Step 1: Representing the Data
For each item, we need to represent the data using features. For example:

- Item features: Category, brand, price range, etc.
- User features: Age, gender, preferences, etc
- User-item interaction: Whether the user likes or dislikes the item (target variable).

Step 2: Computing Probabilities
● Prior Probability: The prior probability is computed for each class (e.g., like or dislike) based on the frequency of each outcome in the training data.
Step 3: Predicting the Outcome
To predict whether a user will like an item, we apply Bayes' theorem to compute the posterior probability for each class (like/dislike) based on the feature values of the item in question. The classifier predicts the class (like or dislike) with the highest posterior probability.
Step 4: Making Recommendations
Once the Naive Bayes classifier is trained, it can be used to predict the likelihood that a user will like a particular item, and this prediction can be used to make recommendations. The system can recommend items that have the highest predicted probability of being liked by the user.

Advantages of Using Naive Bayes for Recommendation Systems
1. Simplicity and Efficiency:
○ Naive Bayes is easy to understand and implement. It is also computationally efficient because it involves simple calculations (multiplying probabilities).
2. Works Well with Small Data:
○ Naive Bayes does not require a large dataset to perform well, making it useful when there is limited interaction data.
3. Handles Categorical Data:
○ It is particularly suited for recommendation systems where the features are categorical, such as item categories, user demographics, and other non-numeric features.

4. Probabilistic Interpretation:
○ Naive Bayes provides a probabilistic interpretation of predictions. It gives the probability that a user will like or dislike an item, which can be valuable for decision-making (e.g., making recommendations based on likelihood).
5. Fast Prediction:
○ Once the model is trained, making predictions is fast, which is useful for real-time recommendation systems.

Conclusion:
In this experiment, the **Naive Bayes classifier** was successfully applied to build a simple recommendation system that predicts whether a user will like or dislike an item based on user characteristics, item attributes, and past ratings. The model was trained on encoded categorical features and demonstrated how probabilistic learning can effectively generate personalized recommendations.